# University of Ottawa

Department of Mathematics and Statistics

MAT 2143 : Introduction to Group Theory

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Final Exam

April 21, 2023

Surname _____ First Name _____

Student # _____ DGD _____

**Instructions:**

(a) You have 180 minutes to complete this exam and submit your PDF file on BrightSpace.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

(g) Cellular phones, unauthorized electronic devices or course notes (unless an open-book exam) are not allowed during this exam. **Phones and devices must be turned off and put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

**By signing below, you acknowledge that you have ensured that you are complying with the above statement.**

Signature _____

Please do not write in the table below.

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Maximum | 5 | 5 | 6 | 5 | 8 | 6 | 5 | 4 | 7 | 3 | 54 |
| Grade | | | | | | | | | | | |

1. (a) **[2 points]** Write down the definition of the greatest common divisor of two integers $m$ and $n$.

**Solution:** See lecture notes

(b) **[3 points]** Determine all integers $n > 2$ such that $\gcd(n^2 - n - 2, n^3 - 2n^2) = 1$. Justify your answer.

**Solution:** We have $n^2 - n - 2 = (n-2)(n+1)$ and $n^3 - 2n^2 = n^2(n-2)$. Thus $n-2$ is always a common divisor of $n^2 - n - 2$ and $n^3 - 2n^2$. Consequently, we should have $n - 2 = \pm 1$. This yields $n = 3$ (note that $n = 1$ is not acceptable since it is assumed that $n > 2$).

2. (a) [**4 points**] Determine all of the elements $x \in U(115)$ such that $x = x^{-1}$.

Recall that $U(n)$ denotes the group of (multiplicative) invertible elements of $\mathbb{Z}_n$.

**Solution:** We have $115 = 23 \times 5$. Now the relation $x = x^{-1}$ is equivalent to $x^2 = 1 \bmod 115$. The latter relation is equivalent to

$$115|(x^2 - 1).$$

To determine all possible values of $x$ we have to consider 4 cases:

- $115|x - 1$. Hence $x = 1$.
- $115|x + 1$. Hence $x = 114$.
- $5|x-1$ and $23|x+1$. From the second constraint it follows that the possibilities for $x$ are $x \in \{22, 45, 68, 91, 114\}$ and since we also have $5|x - 1$ the only possibility is $x = 91$.
- $5|x + 1$ and $23|x - 1$. From the second constraint it follows that the possibilities for $x$ are $x \in \{1, 24, 47, 70, 93\}$ and since we also have $5|x + 1$ the only possibility is $x = 24$.

Thus the final answer is $x \in \{1, 24, 91, 114\}$.

(b) [**1 point**] Is the set $S := \{x \in U(115) : x = x^{-1}\}$ a subgroup of $U(115)$? You should justify your answer.

**Solution:** Yes, we can prove this by the subgroup test: given $x, y \in S$ we have

$$(xy^{-1})^{-1} = (y^{-1})^{-1}x^{-1} = yx^{-1} = y^{-1}x = xy^{-1}.$$

It follows that $xy^{-1} \in S$. Thus by the Concise Subgroup Test we obtain that $S$ is a subgroup of $U(115)$.

3. (a) **[3 points]** Write down the proof of the following assertion:

Let $G$ be a group and let $a \in G$ be such that $o_G(a) < \infty$. Set $n := o_G(a)$. Then for every integer $k \in \mathbb{Z}$ we have $a^k = e$ if and only if $n|k$.

**Solution:** See lecture notes.

(b) **[3 points]** Let $G$ be a group such that $|G| = 715 = 5 \times 11 \times 13$. Also, let $a \in G$ be an element such that $a^2 \neq e$ (where as usual $e$ denotes the identity element of $G$) and $a^{25} = a^5$. Determine $o_G(a)$. You should justify your answer.

**Solution:** Set $d := o_G(a)$. From the relation $a^{25} = a^5$ it follows that $a^{20} = e$. It follows that $d|20$. But also $d|715$ by Lagrange's theorem. Consequently, $d|\gcd(715, 20) = 5$. Thus $d = 1$ or $d = 5$. But from $a^2 \neq e$ it follows that $a \neq e$ hence $d > 1$. In conclusion, $d = 5$.

4. (a) **[2 points]** Write down the definition of the centralizer $C_G(a)$ of an element $a \in G$.

**Solution:** See lecture notes

(b) **[3 points]** Let $\sigma \in S_6$ be the permutation $\sigma := \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 5 & 3 & 6 & 1 \end{pmatrix}$. Also let $\tau := (5\ 1\ 4)(2\ 3\ 6)$; thus $\tau \in S_6$ is a product of two 3-cycles . Does $\tau$ belong to $C_{S_6}(\sigma)$? You should justify your anwser.

**Solution:** We have $\sigma = (1\ 2\ 4\ 3\ 5\ 6)$. Set $\tau := (5\ 1\ 4)(2\ 3\ 6)$. Now by a direct calculation we have

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 1 & 6 & 2 & 4 \end{pmatrix} = \tau\sigma.$$

It follows that $\tau \in C_{S_6}(\sigma)$.

5. (a) **[2 points]** Write down the definition of an odd permutation.

**Solution:** See lecture notes.

(b) **[4 points]** Let $\sigma \in S_6$ denote the 6-cycle

$$\sigma = (6 \;\; 5 \;\; 4 \;\; 3 \;\; 2 \;\; 1).$$

Let $H = \langle \sigma \rangle$ denote the subgroup of $S_6$ that is generated by $\sigma$. The group $H$ contains a unique element $\tau$ such that $o_H(\tau) = 2$. Compute $\tau(3)$. You should justify your answer.

**Solution:** We have $o_{S_6}(\sigma) = 6$. Thus

$$H = \{\varepsilon, \sigma, \sigma^2, \sigma^3, \sigma^4, \sigma^5\}.$$

Since $o_H(\sigma) = 6$, we have

$$o_H(\sigma^i) = 6/\gcd(6, i) \text{ for } i = 1, 2, 3, 4, 5.$$

It follows that the unique element of order 2 in $H$ is $\sigma^3$. Also we have $\sigma^3(3) = 6$.

(c) **[2 points]** Let $H$ be as in part (b). Determine the number of elements of $H \cap A_6$. You should justify your answer.

**Solution:** The 6-cycle $\sigma$ is odd. Also $H = \{\epsilon, \sigma, \sigma^2, \sigma^3, \sigma^4, \sigma^5\}$. Since $\sigma$ is odd, the permutation $\sigma^k$ is even precisely for $k$ even. Thus

$$H \cap A_6 = \{\varepsilon, \sigma^2, \sigma^4\}.$$

6. Let $G := \mathbb{Z} \times \mathbb{Z}$. Also set

$$H := \{(a, a) \ :, a \in \mathbb{Z}\} \quad \text{and} \quad K := \{(a, -a) \ : \ a \in \mathbb{Z}\}.$$

(a) [**4 points**] Using the *concise subgroup test* prove that $H$ and $K$ are subgroups of $G$.

**Solution:** Given $(a, a)$ and $(b, b)$ in $H$ we have $(b, b)^{-1} = (-b, -b)$ hence

$$(a, a)(b, b)^{-1} = (a, a) + (-b, -b) = (a - b, a - b) \in H,$$

as the right hand side also has equal components. The argument for $K$ is similar.

(b) [**2 points**] Does there exist an element $(a, -a) \in K$ such that we have equality of cosets $(a, -a) + H = (37, 43) + H$? You should justify your answer.

**Solution:** The two cosets are equal precisely when $(37, 43) - (a, -a) \in H$, i.e., $37 - a = 43 + a$. This yields $a = -3$. Hence $(-3, 3) + H = (37, 43) + H$.

7. (a) **[2 points]** Write down the definition of a normal subgroup of a group $G$.

**Solution:** See lecture notes.

(b) **[3 points]** Let $G = D_6$. Recall that the dihedral group $D_6$ has two generators $r$ and $s$ such that $o_G(r) = 6$, $o_G(s) = 2$, and $srs = r^5$. Is the subgroup generated by $r^2$ a normal subgroup of $G$? You should justify your answer.

**Solution:** Set $H := \langle r^2 \rangle$. We prove that $H$ is a normal subgroup of $D_6$. Since $r$ and $s$ generate $G$, it suffices to check that

$$rHr^{-1} \subseteq H \quad , \quad r^{-1}Hr \subseteq H \quad , \quad sHs^{-1} \subseteq H \quad , \quad s^{-1}Hs \subseteq H.$$

Note that $s = s^{-1}$ so the last two inclusions are logically equivalent.

We have $H := \{e, r^2, r^4\}$. Clearly $rHr^{-1}$ and $r^{-1}Hr$ are subsets of $H$ (since $r(r^2)r^{-1} = r^{-1}(r^2)r = r^2$). Furthermore

$$sr^2s^{-1} = sr^2s = (r^5)^2 = r^{10} = r^4 \in H,$$

and

$$sr^4s^{-1} = sr^4s = (r^5)^4 = r^{20} = r^{6\times 3+2} = r^2 \in H.$$

These relations prove that $sHs^{-1} \subseteq H$.

8. [**4 points**] Write down the proof of the following assertion:

Let $\phi : G \to H$ be a group homomorphism. Then $\phi$ is an injection if and only if $\ker \phi = \{e\}$ where $e$ denotes the identity element of $G$.

**Solution:** See lecture notes.

9. In this problem let $\phi : \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_3 \to \mathbb{Z}$ be defined by $\phi(a, b, c) := 3a - 2b$.

(a) **[3 points]** Prove that $\phi$ is a group homomophism.

**Solution:** We have

$$\begin{aligned}\phi(a, b, c) + \phi(a', b', c') &= (3a - 2b) + (3a' - 2b') \\ &= (3a + 3a') - (2b + 2b') = \phi(a + a', b + b', c + c'),\end{aligned}$$

hence $\phi$ is a group homomorphism.

(b) **[2 points]** Let $H := \{(x, -x, y) : x \in \mathbb{Z}, y \in \mathbb{Z}_3\}$. We consider the restriction of the map $\phi$ of part (b) to a group homomorphism $\phi : H \to \mathbb{Z}$. Compute $\ker \phi$ explicitly. You should justify your answer.

You do not need to prove that $H$ is a subgroup of $G$.

**Solution:** $\phi(x, -x, y) = 0 \Rightarrow 3x - 2(-x) = 0 \Rightarrow 5x = 0 \Rightarrow x = 0$. Furthermore, it is easy to check that elements of $H$ of the form $(0, 0, y)$ for $y \in \mathbb{Z}_3$ are in $\ker \phi$. Thus

$$\ker \phi = \{(0, 0, y) : y \in \mathbb{Z}_3\}.$$

(c) **[2 points]** Set $L := \phi(H) := \{k \in \mathbb{Z} : k = \phi(v) \text{ for some } v \in H\}$. The quotient group $\mathbb{Z}/L$ is isomorphic to $\mathbb{Z}_d$ for some $d \in \mathbb{N}$. Determine the value of $d$. You should justify your answer.

**Solution:** We have $\phi(x, -x, y) = 3x - 2(-x) = 5x$ for all $x \in \mathbb{Z}$ and $y \in \mathbb{Z}_3$. It is clear that $\phi(H) \subseteq 5\mathbb{Z}$. Furthermore given $t \in 5\mathbb{Z}$, we have $t = 5x$ for some $x \in \mathbb{Z}$ and then $\phi(x, -x, 0) = 3x - 2(-x) = 5x = t$. Hence $5\mathbb{Z} \subseteq \phi(H)$. From this it follows that $L = 5\mathbb{Z}$. Now $\mathbb{Z}/L \cong \mathbb{Z}/5\mathbb{Z} \cong \mathbb{Z}_5$, hence $d = 5$.

10. For each of the following statements, write **T** if the statement is true, and write **F** if the statement is false. You will receive .5 points for a correct answer, and -.25 points for an incorrect answer. (If you leave the answer space blank, you will not receive any negative points.) You do not need to justify your answers.

(a)____ The group $G := S_5 \times U(3)$ has a normal subgroup $H$ such that $[G : H] = 4$.

(b)____ There exists a group homomorphism $\phi : \mathbb{Z}_{100} \to \mathbb{Z} \times \mathbb{Z}$ such that $\ker \phi \neq \mathbb{Z}_{100}$.

(c)____ The matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 5 & 0 \end{bmatrix}$ is in Smith Normal Form.

(d)____ If $p$ is a prime number, then for any pair of subgroups $H$ and $K$ of $\mathbb{Z}_{p^4}$ we have either $H \subseteq K$ or $K \subseteq H$.

(e)____ In the Cayley table of a group $G$, every element of $G$ occurs exactly once.

(f)____ Every proper subgroup of $\mathbb{Z} \times \mathbb{Z}$ is isomorphic to $\mathbb{Z}$.

A proper subgroup is any subgroup other than $\mathbb{Z} \times \mathbb{Z}$ and $\{(0, 0)\}$.

**Solution: T, F, T, T, F, F**